

DataSet1 — Survival Analysis

Survivorship of a census-scored lifespan cohort. Curves are Kaplan-Meier estimates over individuals; individuals unaccounted for at the final census are right-censored unless that policy is switched off in the experiment's configuration.

Experiment type:	Standard Lifespan
Data file:	DataSet1.xlsx
Generated:	2026-08-23 12:43
Factors:	Sex, Density
Treatments:	4
Individuals:	5600
Deaths:	777
Censored:	4823 (86.1%)
Exclusion group:	none
Censoring policy:	unaccounted individuals censored

Omnibus log-rank: treatments differ ($p = <0.0001$).

Experiment summary

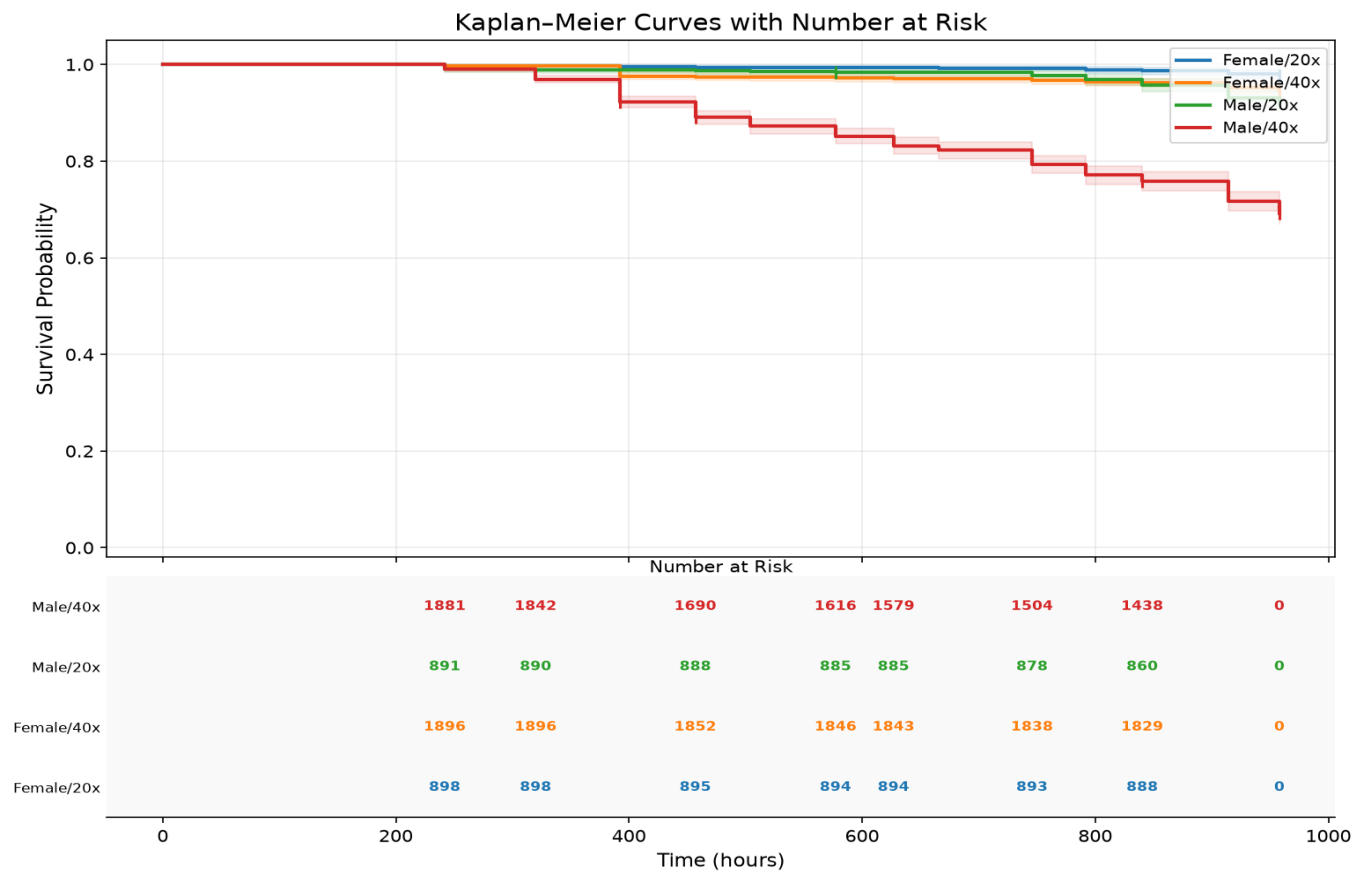
Per-treatment summary

treatment	n_individuals	n_deaths	n_censored	pct_censored
Female/20x	900	20	880	97.800
Female/40x	1900	102	1798	94.600
Male/20x	900	70	830	92.200
Male/40x	1900	585	1315	69.200

Observation window 241.94–958.05, 200 chamber(s).

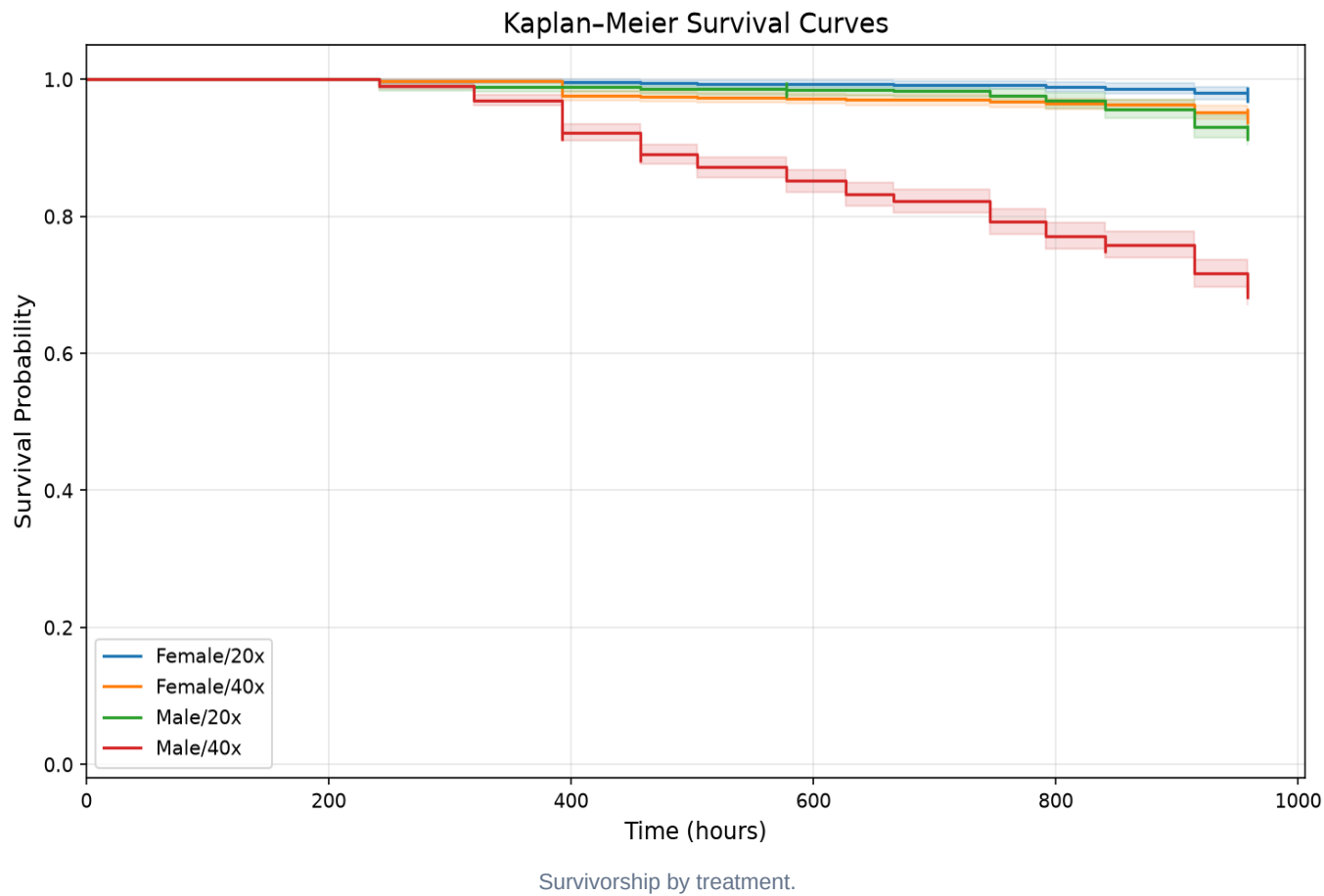
Survivorship figures

KM curves with at-risk table — headline figure

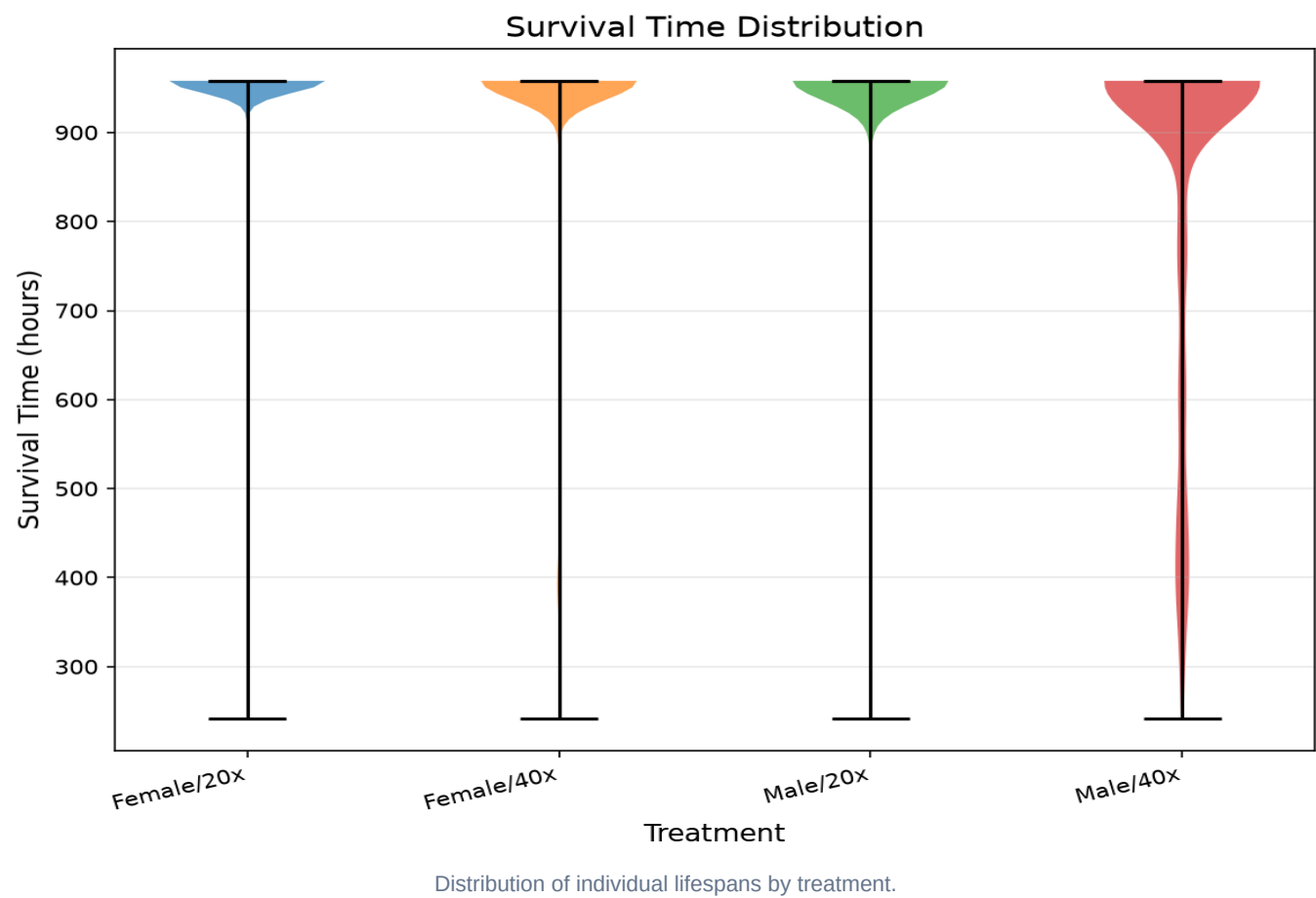


Survivorship with the number at risk beneath the axis.

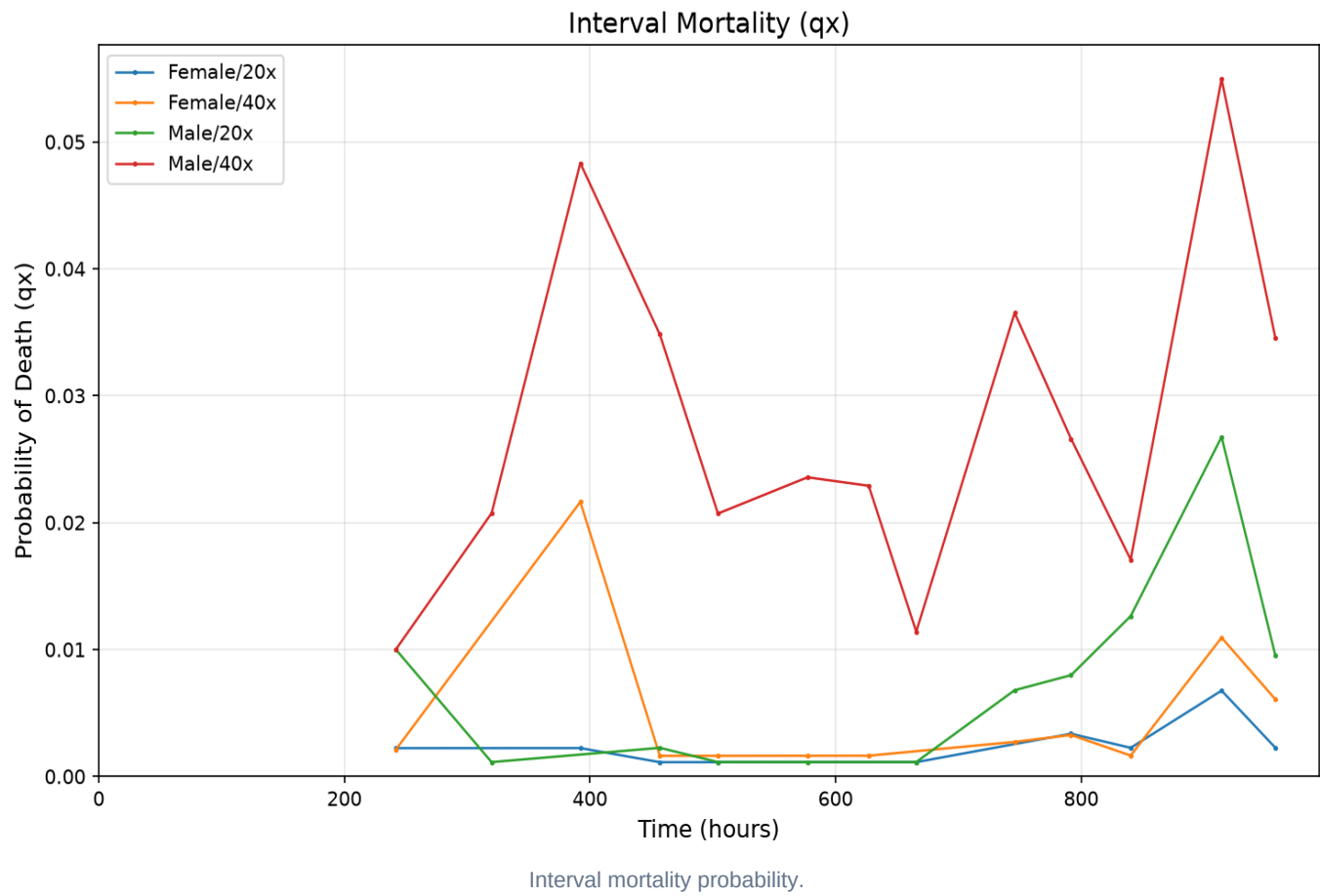
Kaplan-Meier curves



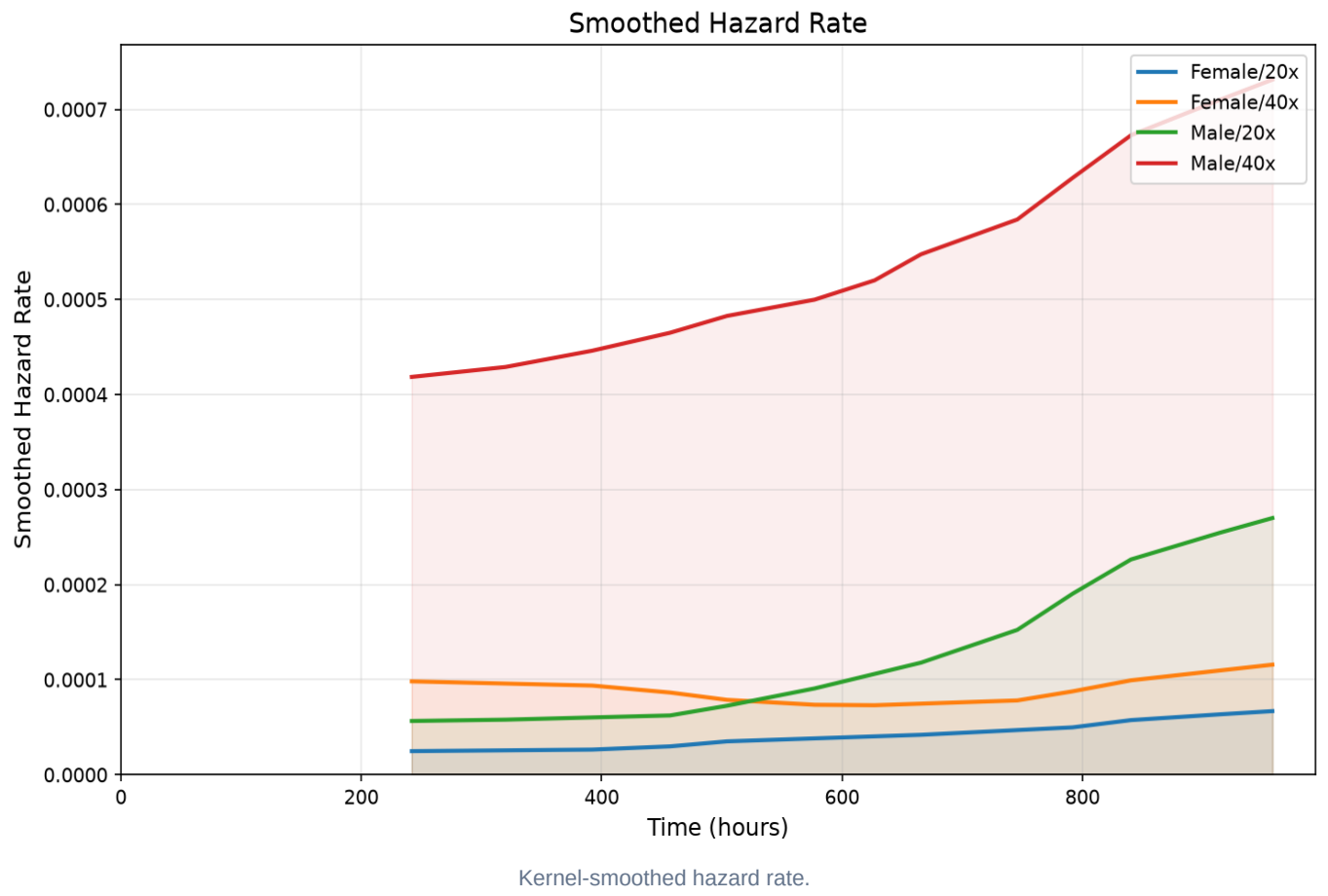
Lifespan distribution



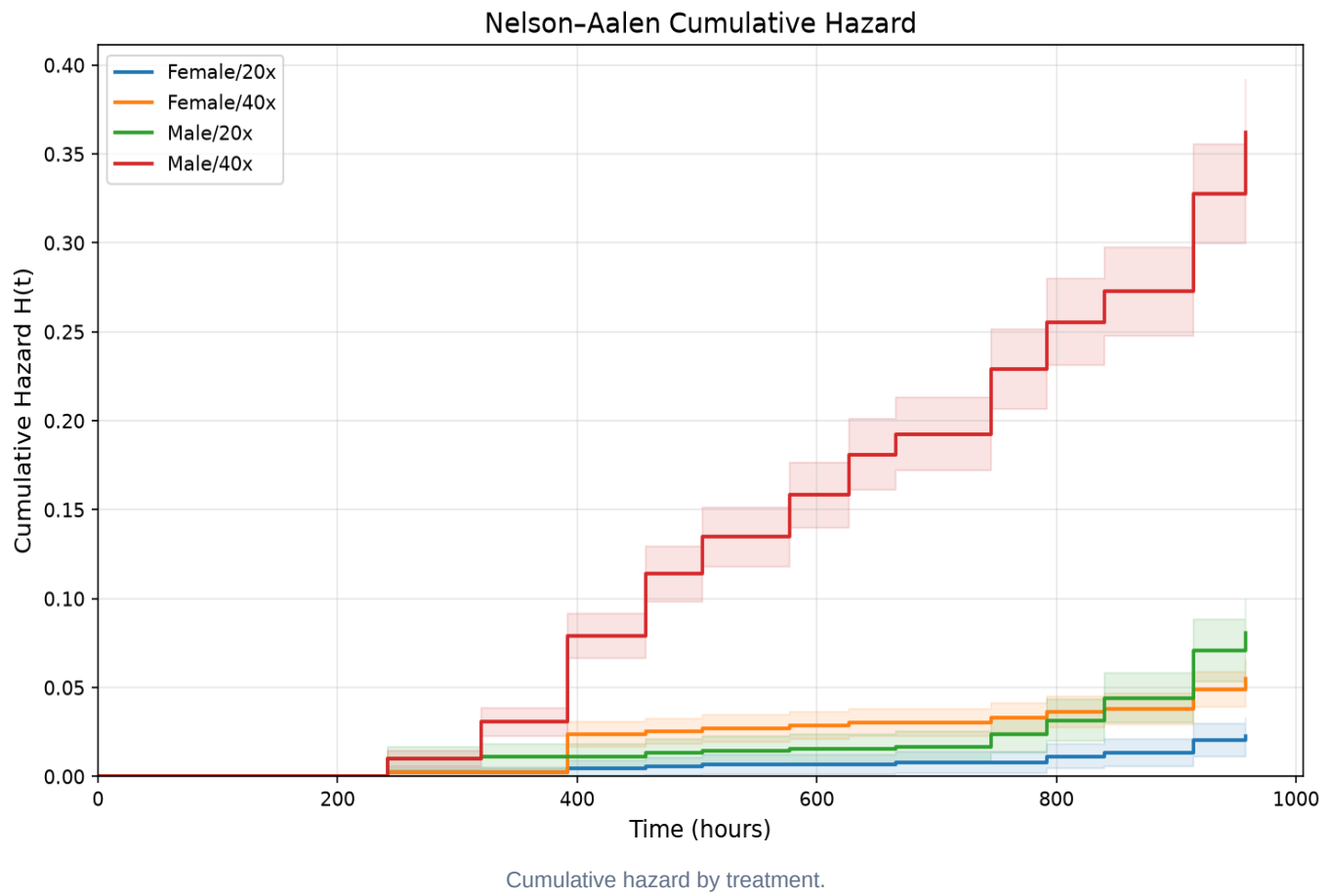
Mortality (qx)



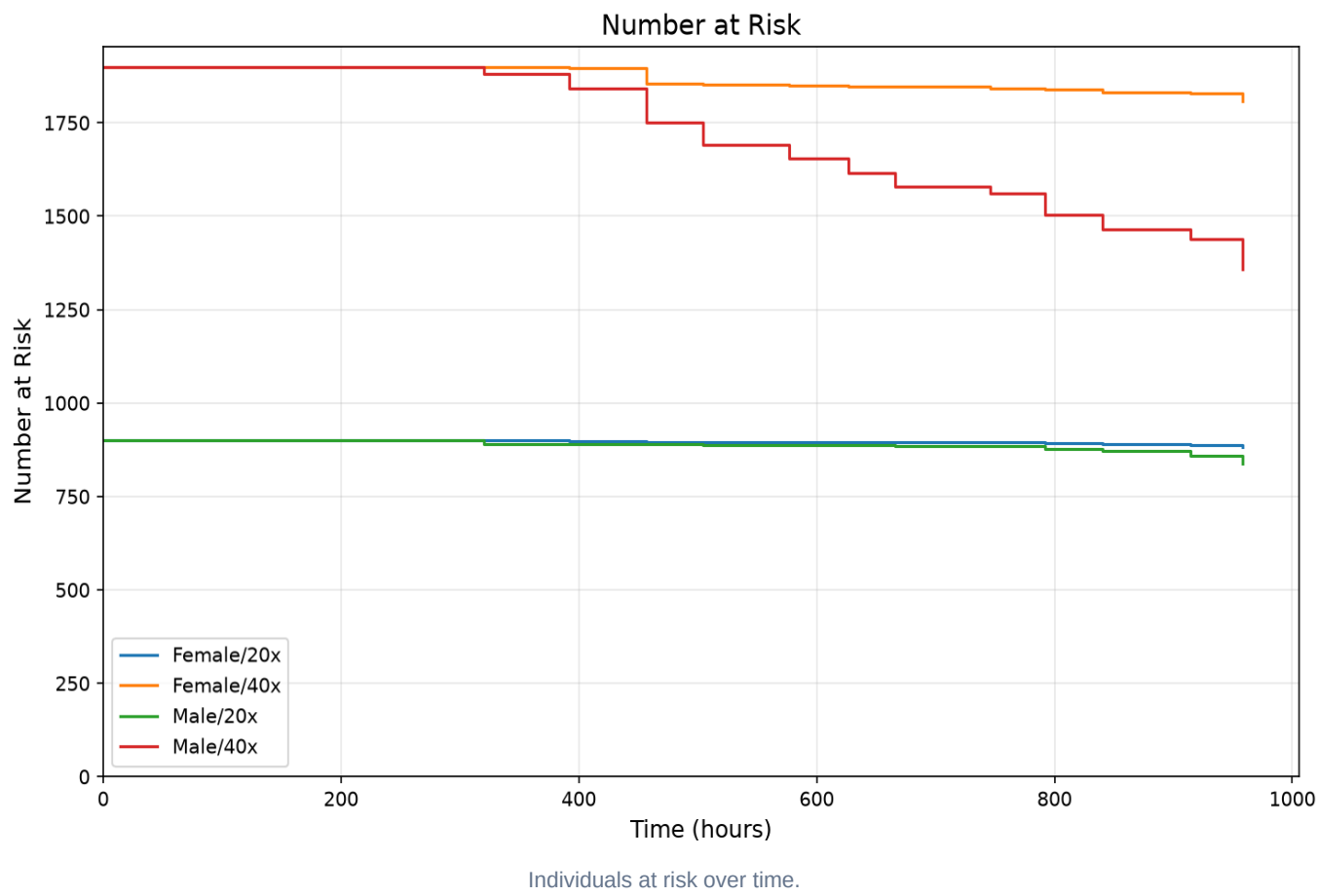
Smoothed hazard



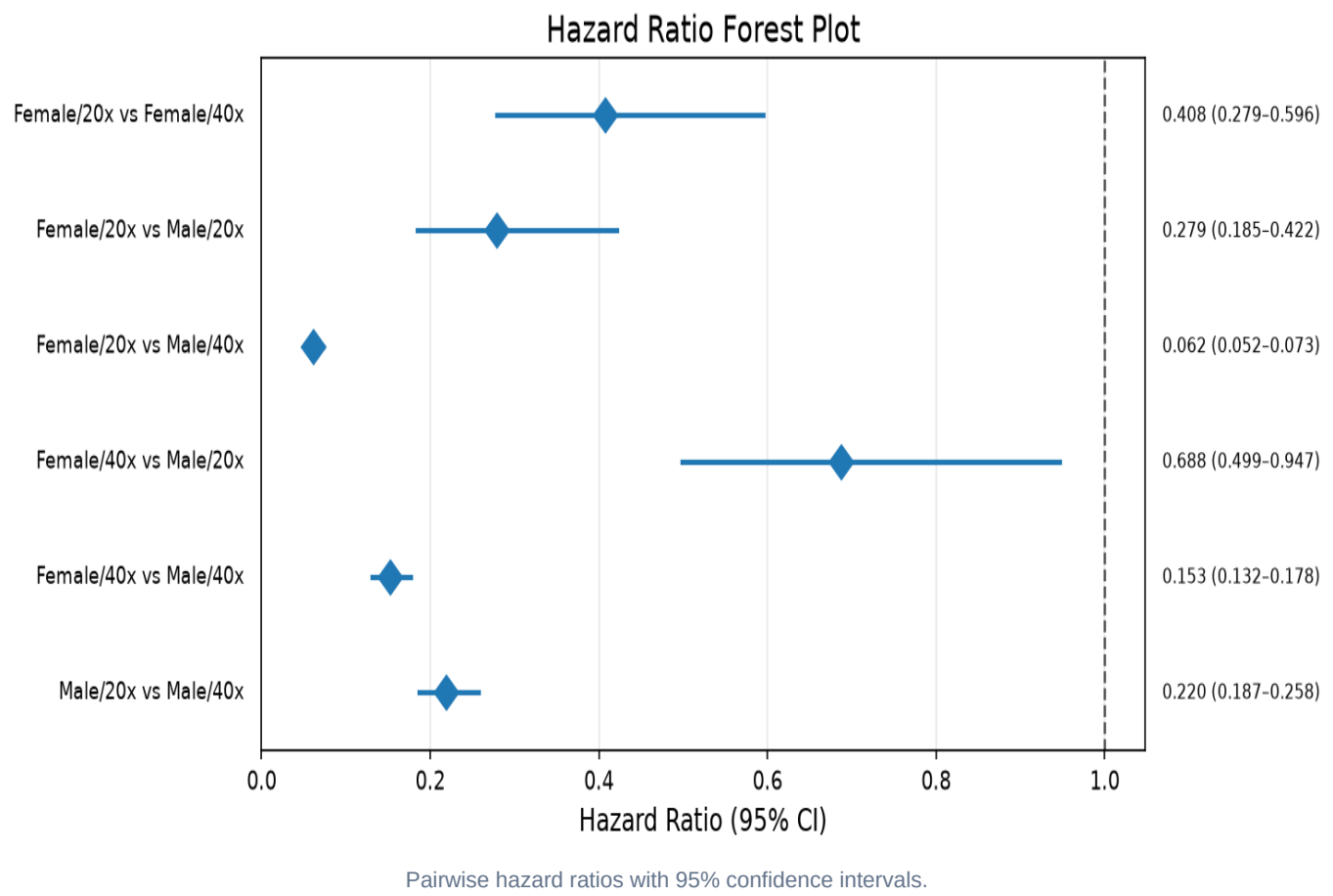
Nelson-Aalen cumulative hazard



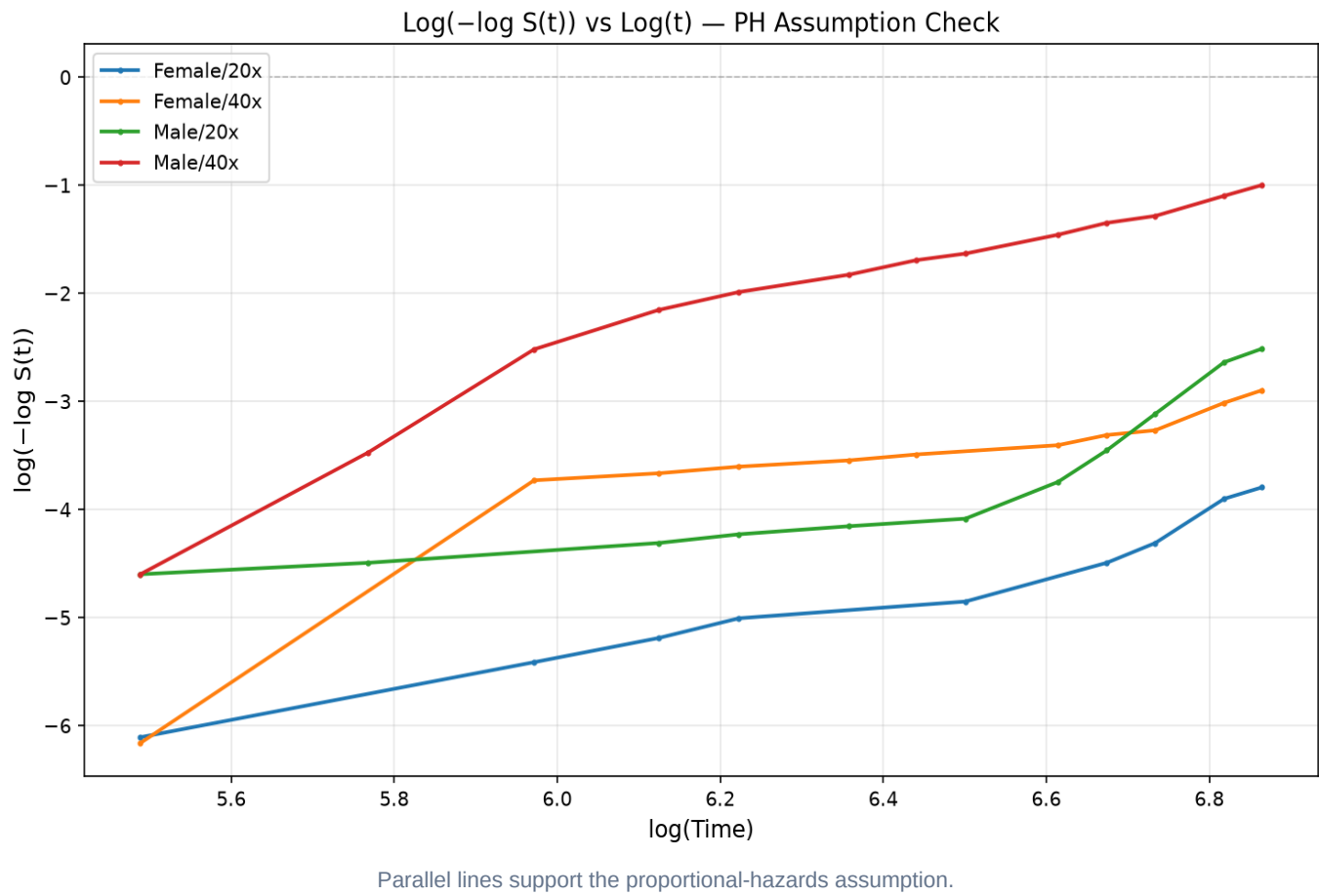
Number at risk



Hazard-ratio forest



Log-log diagnostic



Lifespan statistics

Median survival

treatment	median_survival
Female/20x	—
Female/40x	—
Male/20x	—
Male/40x	—

Mean survival

treatment	rmst	restriction_time
Female/20x	951.565	958.054
Female/40x	937.129	958.054
Male/20x	939.142	958.054
Male/40x	845.827	958.054

Lifespan statistics by treatment

group	n	n_deaths	n_censored	mean_rmst	median	t_max
Female/20x	900	20	880	951.570	—	958.050
Female/40x	1900	102	1798	937.130	—	958.050
Male/20x	900	70	830	939.140	—	958.050
Male/40x	1900	585	1315	845.830	—	958.050

Lifespan statistics by factor level

group	n	n_deaths	n_censored	mean_rmst	median	t_max
Sex=Female	2800	122	2678	941.870	—	958.050
Sex=Male	2800	655	2145	875.860	—	958.050
Density=20x	1800	90	1710	945.530	—	958.050
Density=40x	3800	687	3113	891.940	—	958.050

Survival quantiles

treatment	S=90%	S=75%	S=50%	S=25%	S=10%
Female/20x	—	—	—	—	—
Female/40x	—	—	—	—	—
Male/20x	—	—	—	—	—
Male/40x	456.929	914.349	—	—	—

Survival comparisons

Overall comparison

Test	chi²	df	p	
Omnibus log-rank	740.82	3	<0.0001	***

Pairwise log-rank

group1	group2	observed_1	expected_1	chi2	p_value	df	p_bonferroni	significant_0.05
Female/20x	Female/40x	20.000	39.606	14.491	0.000	1	0.001	True
Female/20x	Male/20x	20.000	45.513	29.176	0.000	1	0.000	True
Female/20x	Male/40x	20.000	215.050	281.304	0.000	1	0.000	True
Female/40x	Male/20x	102.000	116.851	5.943	0.015	1	0.089	False
Female/40x	Male/40x	102.000	365.958	416.488	0.000	1	0.000	True
Male/20x	Male/40x	70.000	231.014	178.010	0.000	1	0.000	True

Pairwise Gehan-Wilcoxon

group1	group2	chi2	p_value	df	p_bonferroni	significant_0.05
Female/20x	Female/40x	14.633	0.000	1	0.001	True
Female/20x	Male/20x	29.099	0.000	1	0.000	True
Female/20x	Male/40x	279.819	0.000	1	0.000	True
Female/40x	Male/20x	5.629	0.018	1	0.106	False
Female/40x	Male/40x	410.245	0.000	1	0.000	True
Male/20x	Male/40x	181.411	0.000	1	0.000	True

Pairwise hazard ratios

group1	group2	hazard_ratio	hr_ci_lo	hr_ci_hi
Female/20x	Female/40x	0.408	0.279	0.596
Female/20x	Male/20x	0.279	0.185	0.422
Female/20x	Male/40x	0.062	0.052	0.073
Female/40x	Male/20x	0.688	0.499	0.947
Female/40x	Male/40x	0.153	0.132	0.178
Male/20x	Male/40x	0.220	0.187	0.258

Data quality

No chambers were excluded from this analysis.

Parametric model fits

Parametric model	AIC
results_by_treatment	—
aic_comparison	treatment model aic log_likelihood median_survival 0 Female/20x Weibull 455.91 -225.9531 3391.31 1 Female/20x Log-Normal 457.19 -226.5957 8071.94 2 Female/20x Log-Logistic 455.97 -225.9848 3836.51 3 Female/40x Weibull 2163.49 -1079.7466 3493.18 4 Female/40x Log-Normal 2159.37 -1077.6827 6607.40 5 Female/40x Log-Logistic 2163.44 -1079.7209 4091.07 6 Male/20x Weibull 1391.35 -693.6738 1858.02 7 Male/20x Log-Normal 1404.55 -700.2753 3095.00 8 Male/20x Log-Logistic 1392.58 -694.2907 2040.04 9 Male/40x Weibull 10135.97 -5065.9874 1270.19 10 Male/40x Log-Normal 10100.90 -5048.4521 1395.00 11 Male/40x Log-Logistic 10130.01 -5063.0069 1334.77
best_model_per_treatment	—